

Lung Cancer Survival Analysis

The objective of the following report is to explore a dataset containing information on the survival time of 137 lung cancer patients with respect to several variables. It will examine which treatments or factors are most influential on the amount of time the patient survives. Completing survival analyses will prove to be useful in the research and treatment of lung cancer. The procedures will be executed using SAS programming.

Exploratory Data Analysis

(1) As with any analysis, the first step is the exploration of data. This will provide an initial understanding of the variables in the dataset, and potentially give insight toward eventual conclusions. The dataset includes the following variables to be analyzed, their type written in parentheses and SAS variable name in italics:

- *treatment* - whether or not the patient received chemotherapy (dichotomous, 2 indicating they have received treatment)
- *type* - cancer type (categorical, four levels)
- *days* - survival time in days (continuous)
- *status* - whether or not survival time is censored (dichotomous, 1 indicating censored)
- *karn* - Karnofsky score, higher numbers representing more severe condition (continuous)
- *diag* - months from diagnosis (continuous)
- *age* - age in years (continuous)
- *therapy* - whether or not the patient received prior therapy (dichotomous, 10 indicating they have received prior therapy)

Summary tables for these variables are shown below, the continuous and categorical variables separated to avoid confusion.

Quantitative Variable Descriptive Statistics							
Variable	Mean	Min	Q1	Median	Q3	Max	Std Dev
days	121.628	1	25	80	144	999	157.817
karn	58.569	10	40	60	75	99	20.039
diag	8.774	1	3	5	11	87	10.612
age	58.306	34	51	62	66	81	10.542

We see that the average number of days survived was a little over 121, and that the variable has a very large range and thus a large standard deviation. Since survival time is the primary measure of interest, a large range may be beneficial as its relationship to the covariates

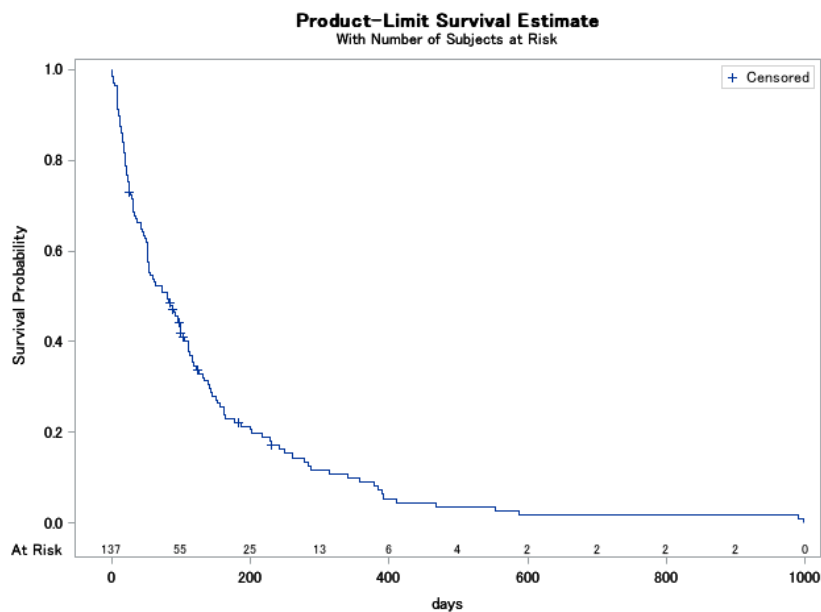
could be better understood. The average Karnofsky score was around 58.5, suggesting that the majority of patients fell into the middle range of condition severity. However, the median and upper quartile were both relatively far from the expected midpoint of 50, so it can be said that the average condition of the patients was fairly dire. Months from diagnosis ranged from 1 to 87, another large range. The quartiles reveal this maximum value to be an outlier, though, which in turn results in a large standard deviation. In any case, this variable may be important to consider as it is generally believed that cancer is more effectively fought if diagnosed and treated early. The last of the continuous variables was age, the average being just over 58. This was potentially predictable as people are typically more likely to develop cancer at older ages. The quartiles for this variable suggest that the vast majority of patients fell between the ages of 50 and 65 years.

Categorical Variable Summary Information					
Variable	Level	Frequency	Percentage	Cumulative Freq	Cumulative Percent
treatment	1	69	50.36	69	50.36
	2	68	49.64	137	100
type	1	35	25.55	35	25.55
	2	48	35.04	83	60.58
	3	27	19.71	110	80.29
	4	27	19.71	137	100
status	0	9	6.57	9	6.57
	1	128	93.43	137	100
therapy	0	97	70.80	97	70.80
	10	40	29.20	137	100

The table above shows the number of patients within each subset of the categorical variables. The treatment groups were almost entirely balanced, with 68 receiving treatment and 69 not. The only reason they are not balanced is the odd-numbered total sample size, but they are sufficiently close to allow for meaningful analysis. The patients were fairly evenly distributed by cancer type as well, although there were many more with type 2 than any of the others. This relative balance will also be useful in the analysis as each type is adequately represented. It can be seen from the summary for *status* that only 9 participants were censored, or lost to follow up in the dataset. This is a very small proportion of the total sample size, meaning there is likely no major concern of the sample size being too small. Lastly, the majority of patients did not receive prior therapy, although nearly 30% of them did. Overall, each of the categorical variables appears to have sufficient representation for all of their respective levels.

Survival Analysis

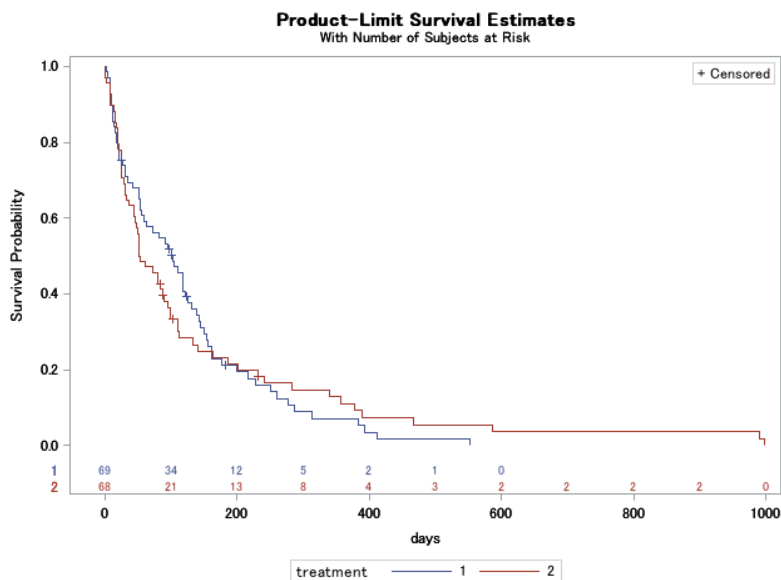
Now that exploratory data analysis has been conducted, the survival analyses themselves can be completed. The first analysis will simply plot the overall survival of patients in the dataset, which will give an initial idea on the deadliness of lung cancer. The Kaplan-Meier curve for this first exercise is shown below.



The curve appears to follow a sort of exponential decay, where many patients pass away after a fairly short period of time, while few of them survive longer periods of time. Specifically, it appears that nearly 90% of the patients have been lost within 365 days. While this shows that lung cancer is especially deadly within the first year, it is not particularly insightful as there is no information about the underlying factors affecting the condition. As

such, the next step will be to analyze whether any of the other variables in the dataset are significant in affecting survival time.

(2a) Several of these other variables could be strongly related to survival time, therefore several analyses will be completed. The first analysis will compare the survival times between patients who received chemotherapy and those who did not. Results could indicate the effectiveness of the treatment on improving the survival time of patients with lung cancer. To clarify, all medians were obtained using SAS output, though the output itself is not included.

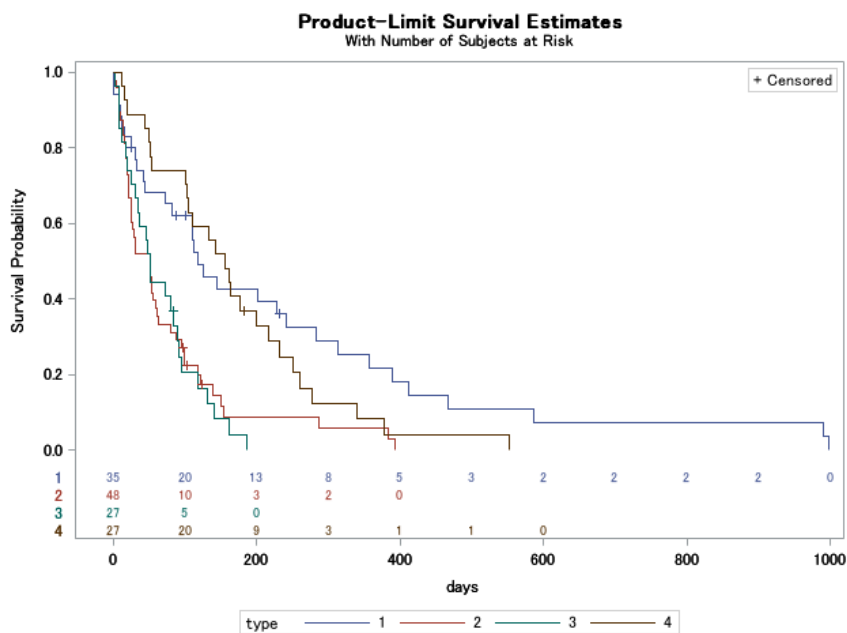


Test of Equality over Strata			
Test	Chi-Square	DF	Pr > Chi-Square
Log-Rank	0.0082	1	0.9277

The curves for the treatment and non-treatment group do not appear to be significantly different. In fact, it shows that non-treated patients actually survive longer

initially, but treated patients ultimately outlast them. Interestingly, the median survival time of the treated group was 52.5, while the median survival time of the non-treated group was 103. The “crossing over” of curves is a major indicator of the treatment having little to no effect, and is supported by the log-rank test whose test statistic was extremely small, meaning the null hypothesis is not rejected and the conclusion is that the survival curves are not significantly different. This conclusion suggests that chemotherapy is not particularly effective at treating lung cancer, despite two treated patients surviving much longer than all others. These results make sense in some respect as chemotherapy is often used as a last resort because of its dangerousness. That is, it is probable that most of the patients receiving the treatment are already in extremely severe condition, making them less likely to survive.

(2b) Since the previous analysis revealed no significant difference between the treatment groups, one might compare the survival times of patients according to cancer type. The following results may indicate which type of cancer is the most severe.

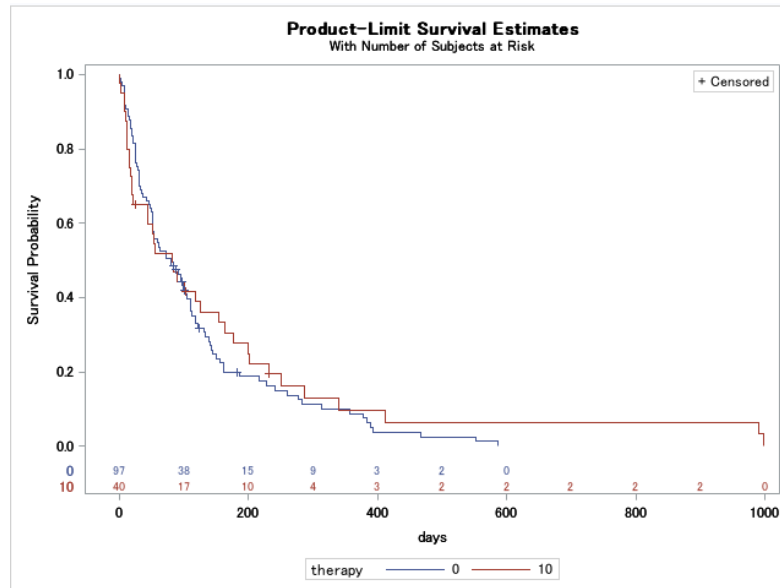


Test of Equality over Strata			
Test	Chi-Square	DF	Pr > Chi-Square
Log-Rank	25.4037	3	<.0001

Adjustment for Multiple Comparisons for the Logrank Test				
Strata Comparison		Chi-Square	p-Values	
type	type		Raw	Tukey-Kramer
1	2	14.8237	0.0001	0.0007
1	3	15.0560	0.0001	0.0006
1	4	0.8739	0.3499	0.7861
2	3	0.4843	0.4865	0.8987
2	4	8.9284	0.0028	0.0149
3	4	7.8476	0.0051	0.0262

The Kaplan-Meier curves seem to illustrate two different “groups” of cancer types, where two separate pairs appear very similar but separated from the other pair. Namely, types 1 and 4 have similar curves, and types 2 and 3 have similar curves. An additional indicator of this is that the medians were as follows: Type 1 = 118, type 2 = 51, type 3 = 51, type 4 = 156. The pairwise comparisons output then supports this, as no significant difference was detected between the curves for types 1 and 4 or between those for types 2 and 3. However, all other comparisons are significant at the 95% confidence level, leading to a significant overall difference between the curves. The output suggests that types 2 and 3 are significantly more severe than types 1 or 4, because patients with these types do not survive as long.

(2c) The third plot will assess the difference in survival times according to the prior treatment of patients. This will provide insight as to whether receiving prior treatment makes a difference in how long a patient will survive.

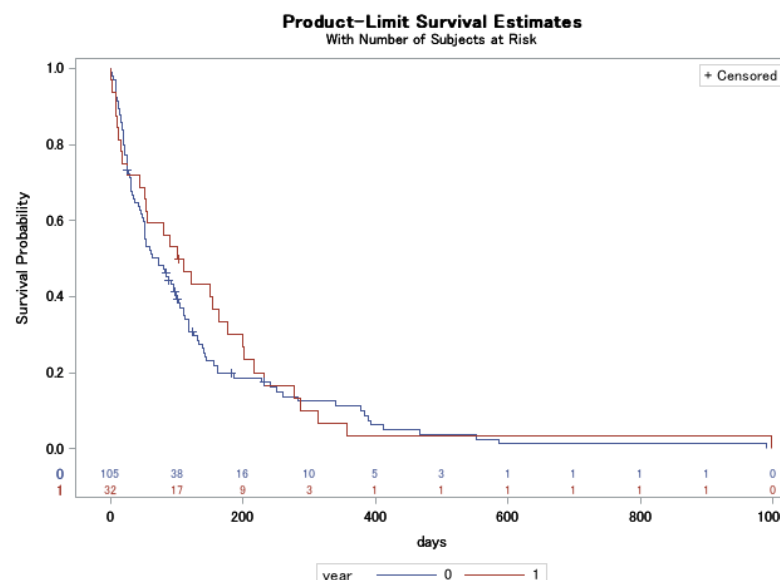


Test of Equality over Strata			
Test	Chi-Square	DF	Pr > Chi-Square
Log-Rank	0.5014	1	0.4789

The curves show that those who have received therapy survive slightly longer than those who have not, though this difference is extremely small and does not apply throughout the duration of the study. The proximity of the curves is reflected by their median survival times, which was 82 days for those who received therapy and 80 for those who did

not. The miniscule difference is again reflected by the Chi-square test on the top right, which gave a very large p-value, indicating no significant difference in survival time due to prior treatment. In context, this means that receiving prior treatment has little to no effect on surviving lung cancer. While this may seem a surprising result initially, there is intuition behind it. If a patient had received prior therapy but was still fighting the cancer, it may be reasonable to assume that the therapy was not effective. Thus, the patient would still have a similar risk of death because the therapy did not reduce any of the risk.

(2d) The following results are from the fourth comparison, which compared the survival time of individuals who had been diagnosed within the previous year to those who had been diagnosed earlier. One might expect those with diagnoses further in the past to have shorter survival times as they have been battling the disease for a longer period of time.

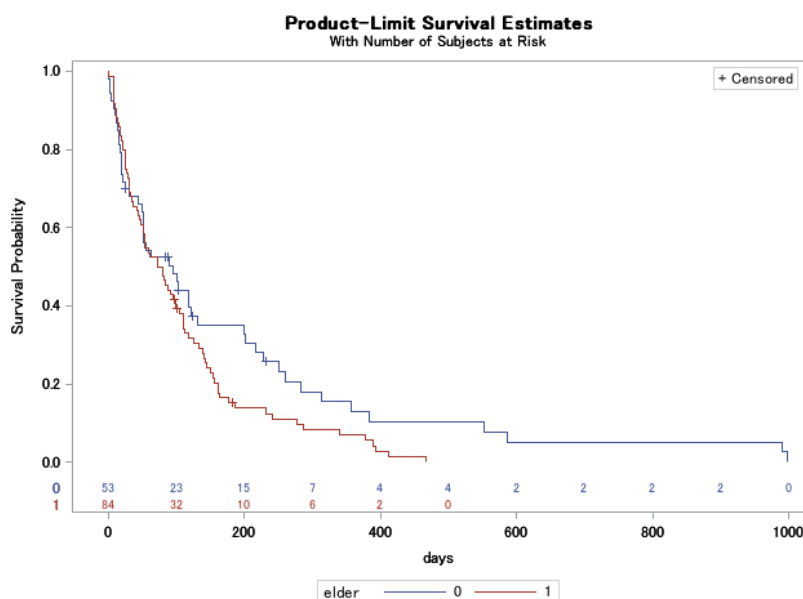


Test of Equality over Strata			
Test	Chi-Square	DF	Pr > Chi-Square
Log-Rank	0.4927	1	0.4827

The median survival time of patients diagnosed within the previous year was 72, while median time for those diagnosed earlier was 105. While this appears

to be a significant discrepancy, the Kaplan-Meier curves appear to be extremely similar once again. Those diagnosed more than 12 months ago initially survived slightly longer, but the curves converge around 250 days, again indicating no difference between the groups. Additionally, the curves for the two groups cross over each other several times, all of which ultimately leading to yet another insignificant result. Hence, the conclusion is that survival time is not significantly affected by whether or not the patient was diagnosed in the previous year. This too seems counterintuitive, because it could be expected for patients diagnosed more recently to survive longer as they have not been fighting the disease for as long. However, there are likely several factors or covariates involved which have led to this result. Such covariates may in fact be the other variables included in the dataset.

(2e) The fifth comparison will stratify using the age variable. Two different groups will be created using a conditional statement in SAS, one consisting of patients aged 60 years or older, and the other consisting of patients aged 59 years or younger. Intuition may suggest that older patients with lung cancer will not survive as long as younger patients, likely due to weakening body systems. The results and analysis are below.

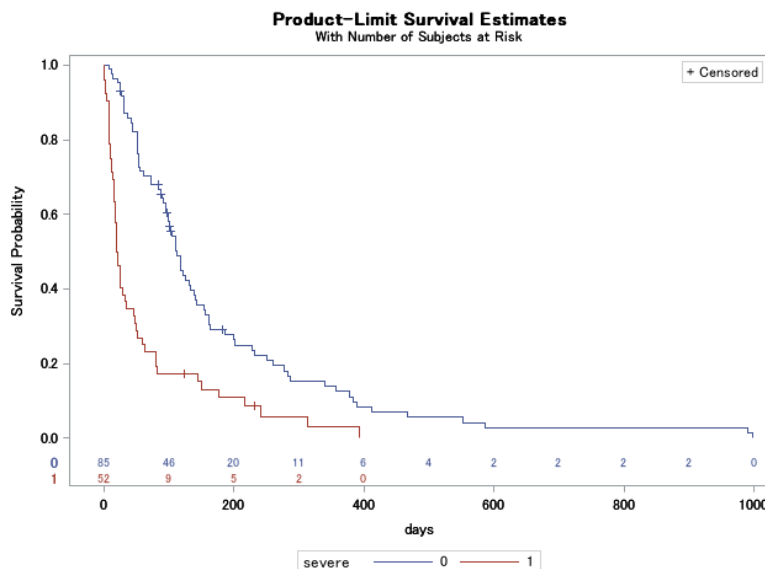


Test of Equality over Strata			
Test	Chi-Square	DF	Pr > Chi-Square
Log-Rank	2.8917	1	0.0890

The curves are initially similar, particularly around the first 100 days of the study. After which, younger patients consistently survive longer, which aligns with the intuition mentioned above. This is reflected by the median survival times, which were 95 for the younger patients and 76.5 for the older patients. The test statistic and corresponding p-value to compare

the two curves, while insignificant for $\alpha = 0.05$, is significant at $\alpha = 0.1$. This p-value likely would have been smaller if not for the initial similarity between the two groups. Regardless, the conclusion of the test is dependent upon the accepted Type I error rate, but the output suggests that age is the most strongly related variable to survival time thus far. It could be said with statistical evidence that the survival time of younger cancer patients is longer than the survival time of older cancer patients, though again this depends on the choice of α . This was somewhat expected due to the aforementioned reasoning that older patients have weakened bodily systems, and are therefore more likely to succumb to cancer.

(2f) The final analyses will determine the relationship between Karnofsky scores and survival time. First, the stratification will use a score of 50 as the cutoff. Afterward, the patients will be stratified into one of following groups of Karnofsky scores: 0-50 (Group 1), 50-80 (Group 2), or 80-100 (Group 3). Generally, a higher Karnofsky score leads to a longer survival time for most illnesses. To specify on the ranges, 0-50 indicates that a patient is in a critical condition and unable to care for themselves, 50-80 indicates that a patient cannot work but is able to complete necessary daily activities, while 80-100 indicates the patient can carry out normal activity and work without special care. The following analyses may provide statistical evidence toward the intuition that lower Karnofsky scores lead to shorter survival times.

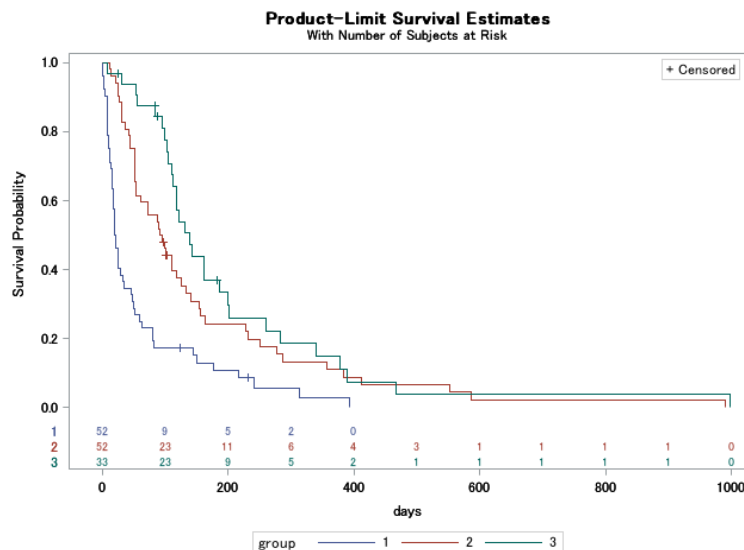


Test of Equality over Strata			
Test	Chi-Square	DF	Pr > Chi-Square
Log-Rank	28.2234	1	<.0001

In this case, the curves are very different from each other. The red curve representing patients with Karnofsky scores higher than 50 is consistently lower and far away from the curve corresponding to patients with lower Karnofsky scores. The median survival times are also very different, with the low group

having a median time of 20.5, and the high group having a median time of 112. The difference is supported statistically by the chi-square test, which yielded an extremely small p-value. Thus, the null hypothesis is rejected and the conclusion is that patients with higher Karnofsky scores survive longer on average than those with lower scores. This result matches the intuition and the purpose of Karnofsky scores.

To analyze the effect of Karnofsky score on survival time further, the patients can be separated into the three groups established in the introduction to this portion. This will provide an even greater understanding on the predictability of Karnofsky scores on lung cancer survival, particularly by revealing which subsets are most different to others. That is, it is entirely possible for scores above 80 to be similar to scores between 50 and 80 in terms of survival time, but for both to be different to scores under 50. Analyzing the three groups using pairwise comparisons will be very useful then, as the precise behavior and effect of Karnofsky score on survival time can be established.



Test of Equality over Strata			
Test	Chi-Square	DF	Pr > Chi-Square
Log-Rank	29.5747	2	<.0001

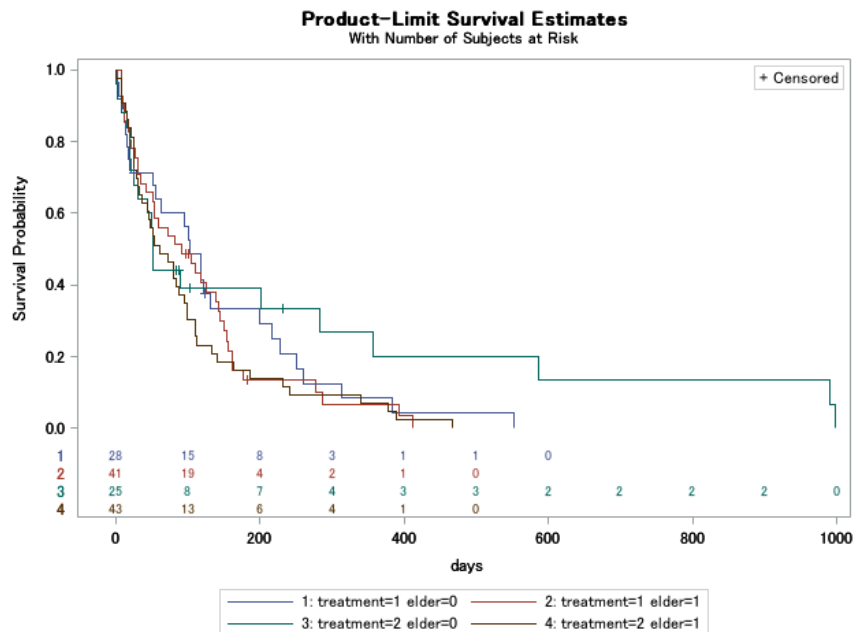
Adjustment for Multiple Comparisons for the Logrank Test				
Strata Comparison		Chi-Square	p-Values	
group	group		Raw	Tukey-Kramer
1	2	13.4569	0.0002	0.0007
1	3	24.1584	<.0001	<.0001
2	3	0.7057	0.4009	0.6781

Groups 2 and 3 appear to be majorly separated from group 1. That is, patients with midrange or high Karnofsky scores have longer survival times on average than those with low Karnofsky scores. The pairwise comparisons reveal that the low score group is significantly different to both of the other groups, but the middle score group is not significantly different to the high score group. The medians were: 20.5 for the first group, 93.5 for the second, and 139 for the third. On its own, this may suggest that the survival time of all three groups are significantly different. However, the Kaplan-Meier curves show that group 3 eventually converges to group 2, resulting in the large p-value seen in the output. In any case the conclusion is relatively unchanged, in that there is significant evidence that lower Karnofsky scores lead to less survival overall than higher Karnofsky scores. Once again this matches the intuition behind the scores, as it is expected for patients in more severe condition to fare worse than those in better condition.

Advanced Survival Analysis

Simple survival analyses have been performed and have found that the most significant variables affecting survival time are Karnofsky score, cancer type, and to a lesser extent patient age. On the other hand, the time since diagnosis and whether or not the patient received chemotherapy *or* prior treatment were found to be insignificant. However, this does not take into account the relationship *between* any of these variables. It is entirely possible that there is some connection between the stratifying variables, and the following analysis will seek to illuminate these connections if they exist.

(3a) The first test will stratify the two treatment groups along with the two age groups, resulting in four survival curves. The results will provide insight as to whether the treatment is more effective for a particular age group, or whether a lack of treatment is particularly bad for one of the groups. The patients were grouped by age according to the previous analysis - those aged 60 years or older and those aged 59 years or younger.

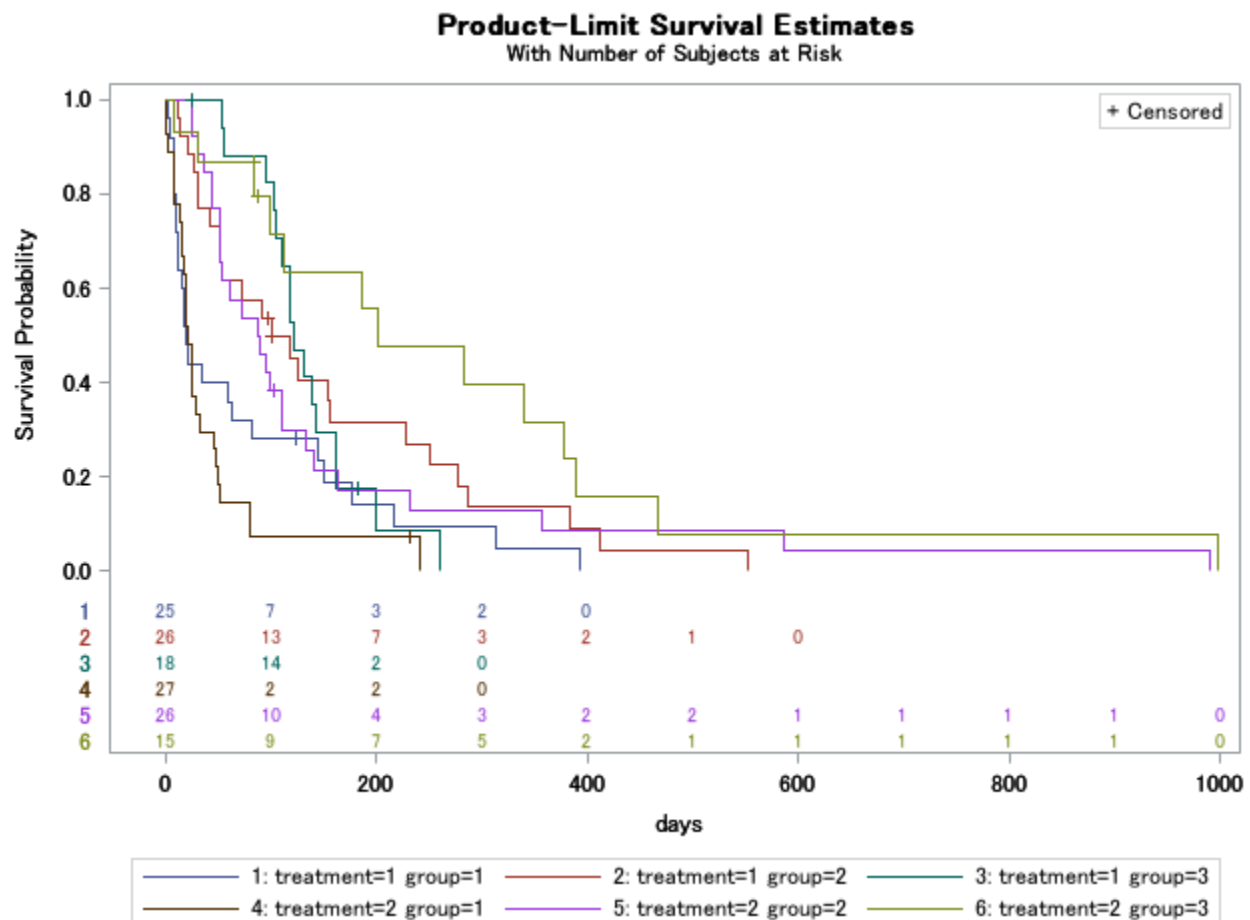


The variable *elder* was equivalent to 1 if the patient was aged 60 years or older, and the variable *treatment* was equal to 2 if the patient was in the chemotherapy group. The curves for each group seem to be very similar to each other for the first 100 days, after which the young treatment group survives much longer on average. The other three groups remain relatively close together in terms of survival time, despite the young non-treatment group

having slightly better survival than either of the older groups. These results are consistent with the Tukey-Kramer comparisons below, in which no p-value indicated a significant difference. The largest difference detected was in fact between the young treatment group and old treatment group, but this p-value was still large, likely because the respective curves cross over. The results may weakly suggest that treatment is slightly more effective for younger patients, which matches intuition as these patients may be better equipped to recover from the harsh chemotherapy treatments than their older counterparts. However, the overall p-value for equality over the strata was 0.2401, meaning there is no significant difference *overall* between any of the curves. It should be noted that the following table format will be truncated to illustrate only the significant or noteworthy comparisons in the coming analyses. This will improve interpretability.

Survival Curve Multiple Comparisons			
Stratum 1	Stratum 2	Chi-Square	adj. p-value
< 60, no treatment	> 60, no treatment	0.1759	0.9752
< 60, no treatment	< 60, treatment	0.5627	0.8767
< 60, no treatment	> 60, treatment	1.5293	0.6034
> 60, treatment	< 60, treatment	1.3127	0.6611
> 60, treatment	> 60, treatment	0.5773	0.8725
< 60, treatment	> 60, treatment	3.9663	0.1911

(3b) The next advanced analysis will stratify by treatment group and Karnofsky score. Since there were three groups of Karnofsky scores, there are six possible combinations of groups and thus six strata. A higher Karnofsky *group* number corresponds to a higher Karnofsky score, thus patients in higher groups can be expected to have longer survival times. The survival plot and pairwise comparisons are once again included below. As mentioned previously, the Tukey-Kramer comparison table will be truncated to include only the significant results, though insignificant results will still be mentioned.

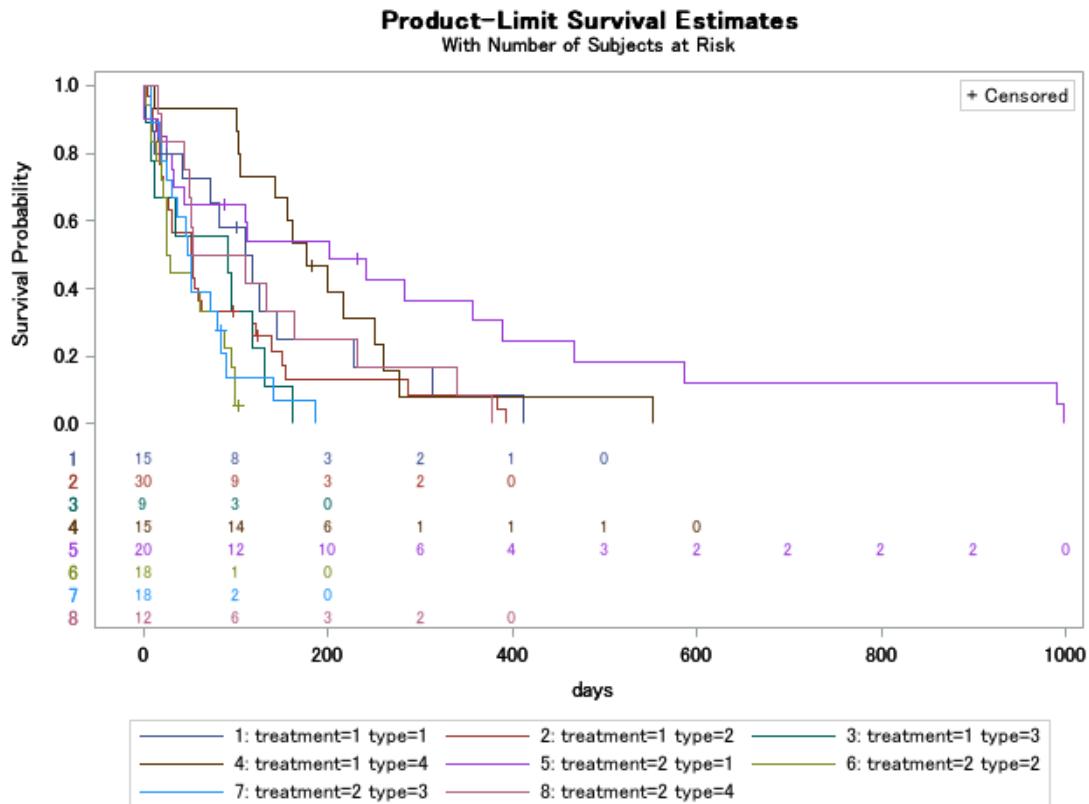


There is clear separation between some of the groups and clear similarity between others. The most apparent difference is between the treated high Karnofsky score group and the treated low Karnofsky score group. All groups in between appear to be relatively similar, and thus are unlikely to be significantly different. Notably, the overall test of equality over the strata gave a test statistic of 38.6848, leading to an extremely small p-value. The null hypothesis is then rejected and the conclusion is that there is *at least one* survival curve significantly different to the others, indicating there is some relationship between Karnofsky score and chemotherapy. The table on the following page illustrates which specific groups were found to be significantly different to each other ($\alpha = 0.05$), omitted comparisons were found to not differ significantly. The significant comparisons will be analyzed in context to give a better understanding of the relationship between Karnofsky score and chemotherapy as they relate to survival time.

Survival Curve Multiple Comparisons			
Stratum 1	Stratum 2	Chi-Square	adj. p-value
Low <i>karn</i> , no treatment	High <i>karn</i> , treatment	11.0441	0.0115
Mid <i>karn</i> , no treatment	Low <i>karn</i> , treatment	12.1292	0.0066
High <i>karn</i> , no treatment	Low <i>karn</i> , treatment	13.0996	0.0040
Low <i>karn</i> , treatment	Mid <i>karn</i> , treatment	9.8010	0.0216
Low <i>karn</i> , treatment	High <i>karn</i> , treatment	25.8507	< 0.0001

It can be seen that the group with low Karnofsky scores that received treatment differs significantly in survival time to nearly every group. This can be seen in the previous survival plot, in which the brown curve is relatively far away from every curve aside from the one corresponding to the non-treated low-score group. The results above are generally consistent with those seen in the analysis of Karnofsky scores alone, mainly because all of the significantly different curves involved a low score. The previous analysis found a low Karnofsky score to be extremely significant in reducing survival time, so it follows that this analysis would find significant differences between low-scoring patients with *any* higher-scoring patients, regardless of treatment. This further supports the conclusion that Karnofsky score is one of the most influential variables affecting the survival of lung cancer. Contextually, these results strengthen Karnofsky scores as a sort of predictor of survival time, a relationship which will be analyzed explicitly in a later section of the report.

(3c) The third comparison will look at the treatment groups against cancer type. With four cancer types, there are eight combinations of groups. This analysis will be useful in that it can provide information as to which type of cancer the treatment is most effective on, the pairwise comparisons giving statistical evidence for more effective treatment. Once again, the Tukey-Kramer pairwise comparisons will be truncated to include only significant results in order to save space and improve interpretability.

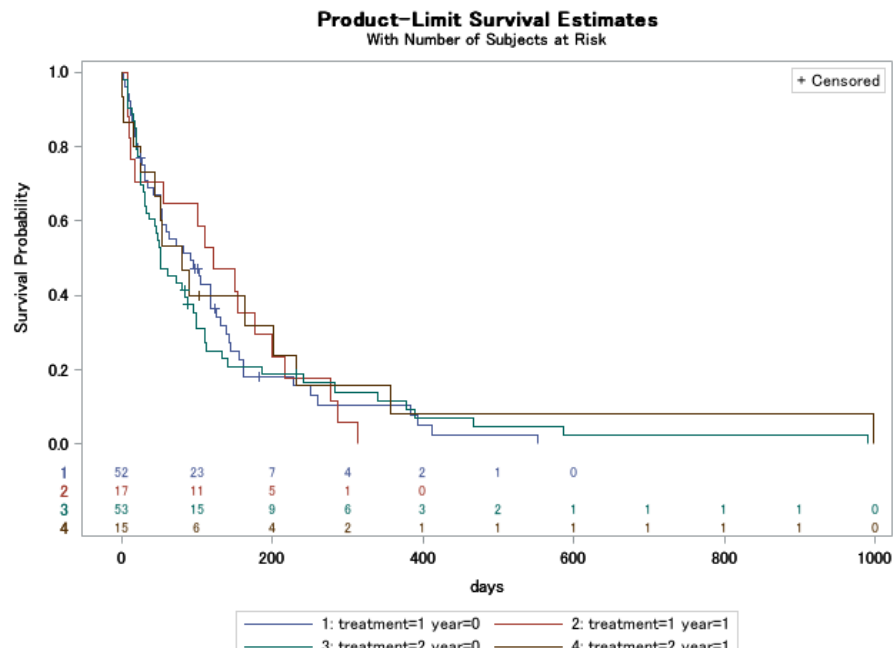


The plot is not particularly easy to interpret because of the number of strata utilized, although there appear to be curves which differ majorly to others. Namely, the treated type 1 cancer group has longer survival time than the other groups, especially the treated type 2 group which diminished to zero in about a tenth the time. To point out the significant comparisons, the following table was created using the SAS output. Interestingly, there were only five significant differences at the 95% confidence level, suggesting that (in tandem) treatment and cancer type only affect survival time in some cases. On the other hand, a number of comparisons had p-values close to 1, indicating virtually no difference in the survival of the groups. Namely the type 1 non-treated group and type 4 treated group, or the type 2 non-treated group and type 3 treated group. This suggests that the variables work to “cancel each other out”, possibly because chemotherapy affects the cancer type, making it more similar to other types which have not been treated. Despite these groups being extremely alike, the overall null hypothesis was still rejected with a p-value of less than 0.0001, meaning there is *some* combined effect of cancer type and chemotherapy on survival. The specific effects will be discussed following the table of results.

Survival Curve Multiple Comparisons			
Stratum 1	Stratum 2	Chi-square	adj. p-value
Type 2, no treatment	Type 1, treatment	9.2904	0.0475
Type 3, no treatment	Type 1, treatment	12.1124	0.0118
Type 4, no treatment	Type 2, treatment	10.8315	0.0223
Type 1, treatment	Type 2, treatment	19.4382	0.0003
Type 1, treatment	Type 3, treatment	15.3005	0.0023

The largest difference detected was between the treated type 1 group and treated type 2 group, with the greatest test statistic and thus smallest p-value. This directly supports the observation on the survival plots, in which the former group survived the longest whereas the latter reached zero in just over 100 days. The results generally suggest that type 1 is far less severe than types 2 or 3, as the patients with it survive much longer on average. This is shown by the first, second, fourth, and fifth rows of the table, and appears to be the case regardless of treatment. However, three of the significant comparisons involved a treatment group being compared to a non-treated group. Exploring this further reveals that treating patients with types 1 or 4 leads to longer survival than *any* patients with types 2 or 3, suggesting that treatment is most effective on the former types. In particular, treatment is especially effective on type 1 cancer, which was found to be significantly different to almost every other potential group aside from type 4 non-treated. Since type 4 non-treated patients and type 1 treated patients have similar survival, one can conclude that cancer types 2 and 3 are much more deadly than types 1 or 4.

(3d) The final area of consideration for the advanced survival analysis is to determine whether time from diagnosis and treatment are strongly related. The connection between these two may be intuitive, because it is generally accepted that catching cancer early and treating it as soon as possible leads to the best results. 12 months will be used as the cutoff point for diagnosis.



Legend for Strata		
Stratum	treatment	year
1	1	0
2	1	1
3	2	0
4	2	1

Test of Equality over Strata			
Test	Chi-Square	DF	Pr > Chi-Square
Log-Rank	0.7678	3	0.8572

Adjustment for Multiple Comparisons for the Logrank Test				
Stratum	_Stratum_	Chi-Square	p-Values	
			Raw	Tukey-Kramer
1	2	0.0107	0.9178	0.9996
1	3	0.1099	0.7403	0.9874
1	4	0.1625	0.6869	0.9779
2	3	0.2795	0.5970	0.9521
2	4	0.1451	0.7033	0.9812
3	4	0.6888	0.4066	0.8403

The survival curves seem to be extremely similar in this case, implying there is no significant relationship between treatment and time since diagnosis on survival time. Many of the curves cross over each other several times, which leads to the results seen on the left. None of the comparisons were significant, so naturally the overall comparison was insignificant as well. The comparison closest to significance was in fact the one hypothesized in the introduction to this sentence. That is, the treated group who had been diagnosed within the previous year differed slightly to the treated group who had been diagnosed earlier. While the latter group survived slightly longer, the difference in survival time was not significant. Overall, these results mean that time since diagnosis and whether or not the patient received chemotherapy are insignificant in affecting survival time, despite the intuition. This

could be due to the fact that many patients may have been battling cancer long before they were diagnosed, so their fight could have been taking place just as long if not longer than the patients diagnosed earlier.

Proportional Hazard Regression

(4) Another method of analyzing the survival time of the patients is proportional hazard regression, which will model survival time as a function of the covariates in the dataset. This will determine which of the variables are most important in explaining the variability in survival time, the result of the modeling can be used to predict new inputs. The first model will include all covariates to gain an initial understanding on the most significant variables, after which a reduced model can be created to improve interpretability while maximizing simplicity.

Analysis of Maximum Likelihood Estimates						
Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio
treatment	1	0.22259	0.18801	1.4017	0.2364	1.249
type	1	0.12926	0.07771	2.7667	0.0962	1.138
karn	1	-0.03514	0.00540	42.3465	<.0001	0.965
diag	1	0.00193	0.00906	0.0455	0.8312	1.002
age	1	-0.00355	0.00916	0.1505	0.6981	0.996
therapy	1	-0.00767	0.02228	0.1185	0.7307	0.992

Several of the parameters estimates were insignificant. Particularly, *treatment*, *diag*, *age*, and *therapy* had very large associated p-values which suggests there is insufficient evidence to conclude the coefficients are significantly different to 0. Hence this indicates that these variables have no significant effect on survival time. Their hazard ratios were mostly close to 1, which is consistent with their insignificant effect because this value means the event is equally likely regardless of the variable value. Treatment had a hazard ratio of 1.249, meaning the instantaneous relative risk of death is slightly higher for patients who received chemotherapy. This result is logical, once again because of the serious side effects of the treatment. Months since diagnosis was estimated to have a positive parameter estimate, suggesting that diagnoses further in the past pose a greater risk of death. Conversely, negative estimates for *age* and *therapy* mean that older patients or those receiving therapy have less risk than their respective counterparts. Older patients having less risk (i.e. a hazard ratio less than 1) is not intuitive, but this result is insignificant meaning the sign of the estimate is effectively null. Though some of these results match intuition, none are reliable because of their lack of statistical significance. On the other hand, cancer type and Karnofsky score were found to be significant, albeit with *type* only significant at the 90% confidence level. The hazard ratio for the former suggests that higher cancer types have greater risk of death, though this does somewhat contradict the interpretation from the advanced survival analysis section. That analysis revealed types 1 and 4 to be less deadly than types 2 or 3, so it makes sense that the risk is generally higher for larger cancer types. However, if the types were listed in ascending order of deadliness (i.e. 1, 4, 2, 3), the hazard ratio would likely be larger and p-value much smaller. The significance of the Karnofsky score comes as no surprise, as it was shown to have a significant effect on survival previously. The negative parameter estimate and hazard ratio of 0.965 suggest that patients with higher Karnofsky scores have less instantaneous risk of death than those with lower scores. This once again matches the purpose of the scores *and* the results of the survival analyses.

Though the previous model was significant in explaining the variability in survival time, it can be simplified because several of its explanatory variables were very insignificant. The following model was fit using the three most significant variables: *treatment*, *type*, and *karn*. The resulting model will ideally be more compact but maintain utility.

Analysis of Maximum Likelihood Estimates						
Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio
treatment	1	0.20633	0.18417	1.2551	0.2626	1.229
type	1	0.12939	0.07763	2.7778	0.0956	1.138
karn	1	-0.03505	0.00515	46.2510	<.0001	0.966

The results of this model were very similar to the previous model. The interpretation of the three variables remains consistent, and the associated p-values for each only changed moderately. It could be noted that the test statistic increased for *type* and *karn*, which would be a result of multicollinearity reduction through removing insignificant variables. That is, the standard error of the estimates are less inflated because there is less redundancy in the

explanatory variables. This also would be a result of these variables “doing more work” in explaining survival time since there are less variables around them to contribute. The most noteworthy change between the models is that treatment became *less* significant because its counterparts became *more* significant. In context, this means that cancer type and Karnofsky score are far more related to survival time than chemotherapy with the given dataset. To clarify the effect of these variables, a hazard ratio of 1.138 for cancer type means that “higher” cancer types are at *greater* risk of death, while a hazard ratio of 0.966 for Karnofsky score means that patients with higher scores are at *less* risk of death. Chemotherapy, though insignificant, had a ratio of 1.229, meaning that patients who received the treatment were at greater risk. The interpretation behind all of these results matches what was discussed in the full model. Notably, this reduced model was still significant overall in explaining the variability in survival time.

The following table was created in order to directly compare the two models. It includes the parameter estimates corresponding to each variable in both the full and reduced models, along with their corresponding p-values in parentheses to indicate significance. The column entitled “-2logL” is the model fit statistic *with* covariates. Lower values of -2logL indicate a better overall model fit.

Model	-2logL	<i>treatment</i>	<i>type</i>	<i>karn</i>	<i>diag</i>	<i>age</i>	<i>therapy</i>
Full	966.222	0.22259 (0.2364)	0.12926 (0.0962)	-0.03514 (<0.0001)	0.00193 (0.8312)	-0.00355 (0.6981)	-0.00767 (0.7307)
Reduced	966.497	0.20633 (0.2626)	0.12939 (0.0956)	-0.03505 (<0.0001)	-	-	-

The full model had a slightly lower -2logL value, but the fit statistics were extremely similar. Since the full model had three very insignificant variables, the reduced model would likely be selected for prediction purposes as it does not sacrifice a considerable amount of fitness and is simpler to use. The reduction of covariates included also led to smaller standard errors of the parameter estimates, making the estimates more useful and their interpretations more reliable. The reduced model would also be more easily utilized for prediction purposes because there are less explanatory variables to keep track of, and thus less concern about any correlation *between* these variables.

Summary

(5) Taking all information into account, it can be concluded that the survival of lung cancer is mostly dependent upon the patient’s Karnofsky score and to a lesser extent the type of cancer with which they are afflicted and whether or not they have chemotherapy. These variables were shown to have the strongest statistical effect on survival time, in particular for Karnofsky score. This was illustrated by both the simple survival analysis *and* advanced analysis in addition to the hazard regression model, all of which found a significant difference in survival times or risk between patients with different Karnofsky scores. In particular, there was overwhelming evidence that patients with lower scores had less survival than those with higher scores, which

directly matches the interpretation of and purpose of the scores in the first place. It is expected for patients with lower scores, i.e. patients in more severe conditions, to pass away sooner than those in less serious conditions. Cancer type was also found to be significant in some respects, particularly that patients with types 2 or 3 are at far greater risk than patients with types 1 or 4, suggesting the former types are significantly more deadly. Treatment, while not overly impactful, was significant in affecting survival time in a few cases. Specifically, if treatment was coupled with cancer types 1 or 4, the patient was far more likely to survive longer than treated types 2 or 3 patients or *any* untreated patients. This means that types 1 and 4 are much more treatable than the other types, making them less deadly *because* of the treatment.

The other three variables: months since diagnosis, whether or not the patient received prior therapy, and age were relatively insignificant. While *diag* and *therapy* were consistently insignificant across all analyses, the effect of age is more difficult to interpret. Its individual survival analysis revealed a nearly-significant reduction in survival time for older patients, but the advanced analysis and hazard regression results suggest it has little to no effect. Since the advanced analysis showed no difference in survival between age and treatment in tandem, and the hazard regression gave a large p-value for the significance of the associated parameter, a logical conclusion is that age is a less significant factor than others. Perhaps the result of the individual analysis is due to the other covariates which were not accounted for.

The proportional hazards regression further supported the relationship between Karnofsky score and cancer type to survival. These variables had by far the most significant effect on instantaneous risk, while the other variables had little or no effect on risk. Additionally, the fact that survival time was able to be represented by half the original number of covariates *without* losing goodness of fit, the effect of Karnofsky score and cancer type is supported even further. The reduced model could also be used to predict the risk of death for a new patient in the dataset because it was overall significant in explaining the variability of survival time. The model could potentially be reduced further to only include Karnofsky score and cancer type, but the inclusion of the treatment variable can still be insightful. Such a model might even be redundant because of the pure significance of Karnofsky scores on survival time. That is, Karnofsky scores appear to be so closely tied to cancer survival that considering other variables may become unnecessary because a confident assertion on survival time can be made with the score alone. With that said, there is certainly still utility in continuing research on the survival of lung cancer, perhaps with the inclusion of new variables or by comparing results to datasets collected in different locations. The strongest conclusions could be drawn from a meta-analysis, in which many studies are taken into account so that all possible information is considered.